



William Pearman

Research Fellow

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 Auckland, New Zealand

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About me

Molecular and microbial ecologist working on population genomics and the evolution of host-microbe (holobiont) relationships.

Scholar Stats

477 citations | h-index: 8 | i10-index: 8

Research Speciality

I am a broadly trained molecular and microbial ecologist with principal expertise in population genomics and microbial ecology. I specialise in developing eco-evolutionary simulations of microbial communities to develop hypotheses which I then test using observational data and experiments. My research background is a combination of population genomics and microbial ecology, with my current research interests working to align these two fields to understand the evolution of host-microbe relationships. My current work combines eco-evolutionary modelling of holobiont evolution, and experimental work with *Daphnia* to determine microbial contributions to host fitness and to understand host-microbial responses to environmental perturbations. In addition to these areas, I am also actively involved in the application and optimisation of new molecular biology protocols for non-model organisms.

Education

2023	PhD Marine Science Dunedin, New Zealand	University of Otago
2020	MNatSci (Distinction) Genetics Auckland, New Zealand	Massey University
2018	BNatSci Genetics Auckland, New Zealand	Massey University

Positions

2024–	Research Fellow Auckland, New Zealand	University of Auckland
2023	Assistant Research Fellow / Lab Manager Dunedin, New Zealand	University of Otago

Distinctions, Grants & Memberships

- 2026 — Visiting Scholar - KU Leuven
- 2025-present — Co-chair - University of Auckland Faculty of Science Research Fellow Society
- 2025-present — Member - University of Auckland Faculty of Science Research Committee
- 2025 — Mana Tūāpapa Future Leader Fellowship (\$820,000 NZD) - Independent 4-year fellowship to explore *Daphnia* microbial ecology
- 2025 — Te Pūnaha Matatini Seed Fund (\$4,880 NZD)
- 2025 — ISME Scholar Mobility Award (\$3,000 NZD)
- 2025 — Emerging Researchers Network - Seed Fund (\$2,500 NZD)
- 2025 — Genetics Society of Australasia - Workshop Support Fund (\$1,000 NZD)
- 2025 — Center for Computational Evolution - Workshop Fund Award (\$3,500 NZD)
- 2025 — Genomics Aotearoa - Workshop Fund Award (\$5,000 NZD)
- 2024 — EEB Representative - University of Auckland Faculty of Science Research Fellow Society
- 2024 — University of Auckland Research Fellow Society Seed Funding (\$5,000)
- 2024 — Invited Speaker - Center for Computational Evolution's 2024 Research Showcase
- 2022 — Best PhD Presentation - New Zealand Molecular Ecology Conference
- 2020 — Best PhD Presentation - New Zealand Molecular Ecology Conference
- 2020 — New Zealand Marine Sciences Society Student Research Award (\$1,500)
- 2020 — University of Otago Doctoral Scholarship (\$27,500)
- 2020 — Invited participant - Ira Moana Investigator Workshop
- 2019 — Invited participant - Ira Moana Project Early Career Workshop
- 2019 — Massey University Masters Scholarship (\$15,000)
- 2018 — Presentation Bursary - Oxford Nanopore Technologies
- 2018 — Massey Graduate Scholarship, Sciences - Massey University (\$5,000)
- 2018 — Natural Sciences Masters Scholarship - Massey University (\$10,000)

- 2017 — Best Student Presentation - New Zealand Molecular Ecology Conference
- 2015 — Vice Chancellors Natural Science Excellence Scholarship, Massey University (\$15,000)

Service & Outreach

- Peer review: Journal of Phycology, Journal of Biogeography, Molecular Ecology, Molecular Ecology Resources

Teaching

- Contributed course material — BIOSCI 761, University of Auckland
- Workshop instructor — SLiM eco-evolutionary simulation workshop
- Workshop instructor (upcoming) — eDNA bioinformatics workshop
- Undergraduate teaching laboratory instructor/demonstrator
- Postgraduate student supervision (multiple students)

Conference Presentations & Invited Talks

- 2024 — Invited talk — Center for Computational Evolution's 2024 Research Showcase
- 2022 — Conference presentation — New Zealand Molecular Ecology Conference (awarded Best PhD Presentation)
- 2020 — Conference presentation — New Zealand Molecular Ecology Conference (awarded Best PhD Presentation)
- 2020 — Invited participant — Ira Moana Investigator Workshop
- 2019 — Invited participant — Ira Moana Project Early Career Workshop
- 2017 — Conference presentation — New Zealand Molecular Ecology Conference (awarded Best Student Presentation)

Publications

19. Ryder, F.J., Pearman, W.S., Layton, C., Parvizi, E., Johnson, C., Bellgrove, A. & Fraser, C.I. (In Press). Contrasting patterns of population structure in two habitat-forming kelp species in southeastern Australia. *Journal of Phycology*. [DOI](#)
18. Kroos, G.C., Fernandes, K., Seddon, P., Ashcroft, T., Pearman, W.S. & Gemmell, N.J. (2026). Targeted Airborne eDNA of an Invasive Wallaby: Effects of Sampler Type, Distance, and Environmental Conditions. *Environmental DNA*, 8(1), e70240. [DOI](#)
17. Pillay, P., dos Remedios, N., Pearman, W.S., Santure, A.W. & Allen, M.S. (2025). Bones, barcodes, and biodiversity: Optimising bulk bone metabarcoding analysis for tropical subfossil collections from Polynesia. *Quaternary International*, 110099. [DOI](#)
16. Pearman, W.S., Rodrigo, A.G. & Santure, A.W. (2025). Within-host microbial selection and multiple microbial generations buffer the loss of host fitness under environmental change. *FEMS Microbiology Ecology*, fiae089. [DOI](#)
15. Morales, S.E., Tobias-Hünefeldt, S.P., Armstrong, E., Pearman, W.S. & Bogdanov, K. (2025). Marine phytoplankton impose strong selective pressures on in vitro microbiome assembly, but drift is the dominant process. *ISME Communications*, 5(1), ycaf001. [DOI](#)
14. Pearman, W.S., Duffy, G.A., Smith, R.O., Currie, K.I., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Host dispersal relaxes selective pressures in rafting microbiomes and triggers successional changes. *Nature Communications*, 15(1), 10759. [DOI](#)
13. Pearman, W.S., Arranz, V., Carvajal, J.I., Whibley, A., Liao, Y., Johnson, K., Gray, R., Treece, J.M., Gemmell, N.J., Liggins, L., Fraser, C.I., Jensen, E.L. & Green, N.J. (2024). A cry for kelp: Evidence for polyphenolic inhibition of Oxford Nanopore sequencing of brown algae. *Journal of Phycology*, 60(6), 1601-1610. [DOI](#)
12. Pearman, W.S., Duffy, G.A., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Long-distance movement dynamics shape host microbiome richness and turnover. *FEMS Microbiology Ecology*, fiae089. [DOI](#)
11. Pearman, W.S., Adams, C.I.M., Monteiro, M., Quesada, A. & Fraser, C.I. (2024). Fine-scale phylogenetic diversity gradients support the Antarctic geothermal refugia hypothesis. *Antarctic Science*, 1-8. [DOI](#)

10. Pearman, W.S., Morales, S.E., Vaux, F., Gemmell, N.J. & Fraser, C.I. (2024). Host population crashes disrupt the diversity of associated marine microbiomes. *Environmental Microbiology*, 26(3), e16611. [DOI](#)
9. Pearman, W.S., Duffy, G.A., Liu, X.P., Gemmell, N.J., Morales, S.E. & Fraser, C.I. (2024). Macroalgal microbiome biogeography is shaped by environmental drivers rather than geographical distance. *Annals of Botany*, 133(1), 169-182. [DOI](#)
8. Thompson, B., Atsawawaranunt, K., Nehmens, M.C., Pearman, W.S., Perkins, E.O., Pipek, P., Rollins, L.A., Tan, H.Z., Whibley, A., Santure, A.W. & Stuart, K.C. (2024). Population Genetics and Invasion History of the European Starling Across Aotearoa New Zealand. *Molecular Ecology*, e17579. [DOI](#)
7. Pearman, W.S., Urban, L. & Alexander, A. (2022). Commonly used Hardy-Weinberg equilibrium filtering schemes impact population structure inferences using RADseq data. *Molecular Ecology Resources*, 22(7), 2599-2613. [DOI](#)
6. Liu, X.P., Duffy, G.A., Pearman, W.S., Pertierra, L.R. & Fraser, C.I. (2022). Meta-analysis of Antarctic phylogeography reveals strong sampling bias and critical knowledge gaps. *Ecography*, 2022(12), e06312. [DOI](#)
5. Fraser, C.I., Dutoit, L., Morrison, A.K., Pardo, L.M., Smith, S.D.A., Pearman, W.S., Parvizi, E., Waters, J. & Macaya, E.C. (2022). Southern Hemisphere coasts are biologically connected by frequent, long-distance rafting events. *Current Biology*, 32(14), 3154-3160.e3. [DOI](#)
4. Pearman, W.S., Wells, S.J., Dale, J., Silander, O.K. & Freed, N.E. (2022). Long-read sequencing reveals atypical mitochondrial genome structure in a New Zealand marine isopod. *Royal Society Open Science*, 9(1), 211550. [DOI](#)
3. Pearman, W.S., Wells, S.J., Silander, O.K., Freed, N.E. & Dale, J. (2020). Concordant geographic and genetic structure revealed by genotyping-by-sequencing in a New Zealand marine isopod. *Ecology and Evolution*, 10(24), 13624-13639. [DOI](#)
2. Arranz, V., Pearman, W.S., Aguirre, J.D. & Liggins, L. (2020). MARES, a replicable pipeline and curated reference database for marine eukaryote metabarcoding. *Scientific Data*, 7(1), 209. [DOI](#)
1. Pearman, W.S., Freed, N.E. & Silander, O.K. (2020). Testing the advantages and disadvantages of short- and long-read eukaryotic metagenomics using simulated reads. *BMC Bioinformatics*, 21(1), 1-15. [DOI](#)